

started successfully. This can be done as depicted in Figure 4.

We can also run the client VM from the *virt-manager* GUI. It is interactive and will guide you through the installation with the given ISO image. After selecting the ISO image, the installation procedure is the same as the installation of the host system as described in our previous article in this series.

Installing the Chef client

The Chef client needs to be installed in the client VM which we created using *virt-manager*. This process of preparing the Chef client is called bootstrapping. We can install Chef client by downloading it from <https://downloads.chef.io/chef-client/redhat/>.

Then, we need to install a tool called Ohai, which is used to synchronise various attributes to the nodes (client) at the start-up of the Chef client. Ohai collects data attributes from platform details, network usage, memory statistics, CPU usage, kernel health-check, host information including FQDN (fully qualified domain names) and statistical information. The private key created in Chef Server should be copied to all Chef clients so that two-way authentication (a handshake) happens between server (public key) and client (private key) when there is any communication between Chef Server and the client.

To download the certificate key from the server to all connected Chef clients, we need to run the *knife* command as follows:

```
#knife ssl fetch
```

This will download the certificate key from the server to the client in the folder */Chef/trusted_certs*. To verify if key authentication between client and server has been successful, we can run the *knife* command in the Chef client node as follows:

```
#knife ssl check
```

```
Connecting to host magdesklinux.wipro.com:443 Successfully
verified certificates from 'magdesklinux.wipro.com'
```

Preparing knife

We have prepared Chef Server (on a Linux guest OS) and the Chef client (on a Linux client), so the next step is to prepare the communication channel between server and client. This is done with the configuration of *knife*, which is the command in the Chef environment to upload cookbooks, create clients, clone configuration between clients, upload and edit roles, and much more. *Knife* can be prepared by running the command as depicted in Figure 5.

As described in Figure 5, you should place the private key (*pem* file) in the */etc/chef-server* folder. Once this configuration is done, we can verify that the handshake of keys is happening and the client is configured properly by executing the following command:

```
#knife client list
```

```
[root@magdesklinux ~]# knife configure
WARNING: knife configuration file found
Where should I put the config file? [/root/.chef/knife.rb]
Please enter the chef server URL: [https://magdesklinux.wipro.com:443]
Please enter an existing username or clientname for the API: [magdes]
Please enter the validation clientname: [chef-validator]
Please enter the location of the validation key: [/etc/chef-server/chef-validator.pem]
Please enter the path to a chef repository (or leave blank):
****
You must place your client key in:
/root/.chef/magdes.pem
Before running commands with knife!
****
You must place your validation key in:
/etc/chef-server/chef-validator.pem
Before generating instance data with knife!
****
Configuration file written to /root/.chef/knife.rb
[root@magdesklinux ~]#
```

Figure 5: Knife command

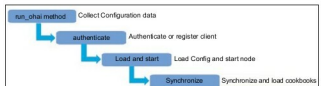


Figure 6: Chef client start-up steps

Now, when we bring up the Chef client, the following basic steps (also shown in Figure 6) happen in the background during client start-up. Please note that these are major steps in the client start-up, though there are many smaller steps involved in between these steps. This stage is explained in Figure 6.

run_ohai: This is the first step when we start the Chef client, where it loads the configuration data as listed earlier. This is collected for each node of the Chef environment and starts the Chef client.

Authenticate/Register: This step is most important to bring up the Chef client into the secured environment of the Chef set-up. If the authentication is successful through the two-way handshake (public/private key authentication), the client node is registered to the Chef Server so that it is entered into the Chef environment and is available in the reporting UI of the Chef management console.

Load and run node: Here, the configuration is loaded, and the client is authenticated and registered to the Chef Server environment. The next step is to load the node and start the node machine.

Synchronizecookbooks: This synchronises the cookbooks from the Chef Server. The cookbooks are stored in the storage of the Chef Server and loaded into the client machines through the *Knife* tool (pushed to all client nodes).

After this, the set-up is finalised. This saves the node state and finalises the client set-up. If any of these steps fail, the Chef client moves to *start-up_run_failed* state and stops abnormally. We need to look into the log file to analyse the reason for the failure and fix it.

Now, run *runs list* to get the list of RUN_ID of the Chef client which is successfully running. This is done by executing the following command:

```
# knife runs list
```

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